SMALL DIFFUSION, BIG IMPACT: COMPOSING IMMERSIVE ACOUSMATIC MUSIC FOR THE NON-IDEAL LISTENING SITUATION

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ABSTRACT

Acousmatic music is increasingly composed with the artistic goal of diffusion through large or complex speaker arrays, over more accessible methods such as the diffusion of mono or stereo works through headphones or other small devices. In this paper, the term "small diffusion" is introduced to describe simpler, often more intimate listening experiences. By awarding a collective label to more portable or accessible methods of acousmatic listening, small diffusion methods are raised to an equal level of legitimacy as more complex diffusion systems in our increasingly technological listening environment. The concept of the non-ideal listening space is argued as a valid and interesting space for acousmatic music. I include two examples of my work composed for small diffusion and non-ideal listening spaces: Paranoia in the Bush and 3 miniatures for headphones. These works engage with the challenge of maintaining spatial and musical immersion despite the possibility of increased distractions and technical obstacles in the small diffusion space, and are thus composed for a larger and more likely range of listening situations to demonstrate a method for increasing access to modern acousmatic music outside of the academic sphere.

1. INTRODUCTION

Music composed for stereo diffusion was a staple of electroacoustic music practice in the 1960s (Austin 2000). Now, acousmatic music is increasingly composed with the artistic goal of spatial immersion, often via large speaker arrays ranging from 5.1, through to Ambisonic arrays using tens of speakers. Composing works for large diffusion systems in concert-style performance is believed to enhance the audience's musical immersion, appreciation of structure, morphology and technique, making it preferable to composing mono or stereo works for accessible playback systems and equipment such as home computers, stereo speakers and headphones (MacDonald 1995). This paper challenges the notion that bigger is always better (immersive, valuable, artistically powerful) when composing acousmatic music. I put forward two new concepts for consideration which have the potential to re-imagine the acousmatic genre for solo listeners and small groups, recalling the great ability of recorded music to bring new musical genres to the wider public (Millard 2005). In this paper, I offer two musical examples to demonstrate these concepts in action: my recent works *Paranoia in the Bush* and *3 miniatures for headphones.* These works employ different compositional and spatial techniques intended for ubiquitous playback tools, such as headphones, to create immersive acousmatic music.

The issue of creating realistic and immersive space in acousmatic works is already quite widely discussed; in particular, the obstacles inherent to stereo and mono diffusion. In "Ambisonics and Acousmatic Space," Natasha Barrett (2010) writes that it is crucial for a listener to place themselves within a space before they can appreciate musical content, suggesting that the discrepancy between real-world spatial perception and that allowed by traditional diffusion systems, i.e. stereo or multi-channel, may be a contributing issue. Alistair MacDonald (1995) similarly asserts that listeners subconsciously judge electroacoustic music by different standards to instrumental music; while one would not question the real-world source of an oboe note, one struggles to relate abstract acousmatic sound to any real-world source object. This arguably makes real-world spatialisation an important tool for engaging the uninitiated. Macdonald (1995) lists multiple issues encountered in stereo speaker diffusion for large audiences: a limited sweet spot; varying distances from the speakers; and a reduced "level of engagement" for those listening from the sides of the stereo image. Mark Nazemi (2014) writes on immersive soundscapes as a method of chronic pain alleviation, and identifies spatial concerns such as differences in Head-related Transfer Function (or HRTF, a result of an individual listener's head, ear and facial formation which affects how sound is perceived and localised) which result in unrealistic sound localisation and hence decreased listener immersion.

It is certainly possible to create realistic headphone spaces through binaural and Ambisonic mixing; these give the impression of sounds occurring from beyond the physical scope of the headphones, as if from a real-world source. The purposeful panning of sounds to be perceived as coming from inside the head is known as in-head acoustic imaging (IHAI). IHAI is defined by Charles Stankievech (2007) as creating an imaginary sounding space within the listener's body. Used artistically by composers such as Ryoji Ikeda and Bernard Leitner, the head cavity is made to sound hollow, while sound objects resonating within this

imaginary space are heard from within by the outside ear (Stankievech 2007). Being quite removed from real-world experience, this unique space eliminates the concern of localisation issues caused by HRTF, inter-aural amplitude and time differences (Stankievech 2007). Nazemi (2014) lists IHAI (which he names the "sound-in-head" phenomenon) as an obstacle to achieving realistic space. Re-framing IHAI as a non-realistic but imaginative spatial technique allows composers to exploit spatial quirks in relatively cheap technologies to create fantastical spaces - which are immersive in their own right.

2. SMALL DIFFUSION

Denis Smalley defines diffusion as "the projection and the spreading of sound in an acoustic space for a group of listeners — as opposed to listening in a personal space" (Austin 2000). Considering this accepted meaning, "small diffusion" may seem like a misnomer. However, diffusion has multiple definitions with slight variations for different contexts; the Oxford Dictionary calls it "the spreading of something more widely." Small diffusion can therefore be taken as a dual term describing not only the physical phenomenon of listening to acousmatic music on portable, accessible diffusion systems, as well as a cultural one: embracing the use of readily available technology as a valuable and legitimate way to make acousmatic (or indeed, any) music.

Due to the nature of affordable technology (such as headphones, CD players, phones, laptops, etc.) the term suggests a listening experience for small groups or individual listeners. While this concept isn't quite the antithesis of the group listening experience, this approach does not require specially built auditoriums or expensive sound apparatus, leading to a natural association with non-specialist or even low-quality equipment, and non-ideal listening conditions such as the presence of external noise, visual distractions, etc. The use of readily accessible equipment to create music is a crucial feature of Ubiquitous Music, which according to Marcelas S Pimenta et al. (2009), "can be defined as musical computing environments that support multiple users, devices, sound sources and activities in an integrated way." Lars Holmquist introduced the term to academic literature in 2005, writing that society lives with access to a "limitless supply of music." Inevitably, this suggests that music will be increasingly "experienced in [physical] isolation," again highlighting how composing for small diffusion might be of benefit for an acousmatic composer hoping to disseminate their work (Holmquist 2005).

However, this boundless portable music supply does not have to be isolating; multiple studies have shown that synchronised movement or dance performed at silent discos results in increased feelings of "social closeness" in listeners (Tarr et al. 2015). Interaction design has progressed to the point that several systems will allow users to "listen in on" others' broadcast devices, which holds (apart from social and security difficulties) great potential for small diffusion listening to become a group experience, while still a deeply personal one (Holmquist 2005).

Nevertheless, small diffusion is a suitable choice for intimate, close listening. Andean (2014) describes the possibility of the listener "hijacking" the composer's intentions with their own unique interpretation of an acousmatic narrative. Art theorist Stephen Scrivener (2002) takes a more pacifist approach, attesting that the end-point of art is not to communicate absolute knowledge, but to encourage the consumer to devise one of many interpretations. Composing for intimate or solo listening (or individual experiences in a group setting, such as the systems described above or Silent Discos) enhances the listener's ability to respond independently and uniquely to the acousmatic. Therefore, it would seem fitting that there should be a name for listening methods which encourage this specifically individual response.

It should be noted that it would be short-sighted to discount the vastly different considerations which make up the evaluation of acousmatic works as compared with the rest of the Western musical canon. The manipulation of space is one of the foremost, or as Smalley (2007) describes it in his seminal work, Space form and the acousmatic image, "aesthetically central." However, that is not to say that small diffusion methods do not retain an important space in electronic art music, not only through its own artistic potential (a discussion which is owed its own paper), but for the sake of accessibility.

2.1. Accessibility

"Accessibility" in this paper refers not to how approachable the acousmatic work is, but to technological accessibility: creating works which are easily diffusible on mediums used by the wider community, in a traditionally and increasingly multi-channel practice. Andean (2012) writes:

> *If one is concerned about making music cross-culturally communicative, there is more urgent need for a bridge between the acousmatic community and the uninitiated.*

Given these observations and critiques, optimising some acousmatic works for easily accessible mediums may be the first step. When first introduced, recorded sound was a "great educator," which allowed the musically curious to learn and pursue music of different cultures and styles (Millard 2005). It encouraged the spread of music which could not be documented by existing Western notation, allowing the development and fusion of multiple genres to take place (Millard 2005). The rapid expansion of affordable, portable music technology has enabled music to be listened to anywhere; hence recorded sound is better able than ever before to disseminate non-notated genres such as electroacoustic music (Holmquist 2005).

One great benefit of using small diffusion methods is portability. A monophonic or stereo image is easily transmitted over radio, CD player, studio monitors, or indeed headphones, which can be found almost anywhere in the world. David Berriman (1998) explains that headphones can be preferable to speakers in many instances; for example, they avoid room and box resonances, hence giving an experience which is truer to the original recording. MacDonald (1995) stresses the importance of space to a listener's interpretation:

> *For even if the composer carefully considers space and spatialisation…as soon as a piece is played in another room, many aspects of the sound are subject to the acoustic qualities of that room.*

Composing for headphones eliminates this concern; the composer who makes music with and for headphones will be able to guarantee quite closely what the listener will hear. Berriman (1998) does concede, however, that headphones may not communicate the same intangible sense of power and bass rumble that speakers will, a lost opportunity for enhanced immersive effect.

3. PARANOIA IN THE BUSH

Since the rise of stereophonic sound in the 1950s, headphones have maintained a steady stake in the listening landscape as ideal vessels for stereophonic listening (Berriman 1998). Therefore, it is in my interest to explore composition for diffusion methods such as stereo speakers and headphones, broadening what MacDonald (1995) boldly describes as the "limited appeal" of electroacoustic music through the accessibility of small diffusion and consideration of likely listening spaces. As an acousmatic composer, I have decided to explore headphone music as a widespread example of small diffusion.

My composition *Paranoia in the Bush* (2019)[1] seeks to represent a variety of sensations experienced while walking alone in the Australian bush – ranging from the frenetic paranoia to meditative tranquility. I exploit IHAI to increase the sensation of uncanny happenings and irrational imagination. The work features sharp cuts between contrasting sounds, frenetic pacing, and occasional moments of calmer, extended field recordings captured at Bundanon Trust, NSW. Without explicitly referencing it, this work was strongly influenced by Bernard Leitner's *Kopfraume* album (2003), which specifically showcases IHAI through simple examples like hand-beaten drums.

No binaural spatialisation was necessary for this work; detailed manual automation of the Standard Panner parameter in Cubase (set to Equal Power pan law) was used to achieve IHAI. As with all acousmatic works, it was necessary to balance a demonstration of spatial characteristics, restraint and subtlety to avoid aural and mental fatigue. A sense of distance and resonance is achieved through a combination of reverbs, panning and EQ; though rather than perceiving sound as far away from the body, the listener may perceive the sound as being far on the perimeter of the sounding space which is inside of them - yet somehow larger than reality.

To enhance the immersive performative experience of this work at its premiere (at a bustling experimental music concert), the listener was invited into a small camping tent decorated with string lights and a bean bag seat, with noise-cancelling headphones provided to listen to the piece.

4. THE NON-IDEAL LISTENING SPACE

By carving out an individual space apart from the existing environment and eliminating external noise, visuals and nosy concertgoers (all potentially distracting), the listener is presented with the *most* ideal listening space for this work.

The Non-Ideal Listening Space (NILS) is a subjective concept whose definition may vary for each composer. A general definition of a NILS might be: a listening situation which poses a risk of distraction to the listener, or non-ideal conditions for performance of a work. Equally, the NILS might mean any scenario outside of a concert hall, studio, or certain site that a work is composed for. Distraction can come in many forms: sensory, sound, light, mental or physical discomfort, and so on.

While small diffusion and the NILS space are two discrete concepts, they often co-exist. Due to the portable nature of small diffusion mediums, it is likely that small diffusion works will be listened to in a NILS. It therefore seems logical for composers to engage with this possibility in the creation and performance of recorded works, no matter what diffusion system their work is intended for. Composers engaging in the medium of small diffusion, in particular, may react to the NILS in two ways:

- By embracing it, and incorporating aspects of the NILS into their work. This may include site-specific performances, Sound Walks, or works which are intended to mingle or react to their sound/physical environment.
- By acknowledging the potential of a NILS and attempting to eliminate its possible distractions. This may include immersing the listener in a *different* sound space; the use of noise cancelling headphones and physical segregation at the premiere of *Paranoia in the Bush* is one example.

It may seem easier to describe a NILS than it does an ideal listening space; even when every care is taken, distraction is often out of the composers' hands, or requiring "technically complex solutions" (MacDonald 1995). As Denis Smalley is quoted by MacDonald (1995):

[1] *Paranoia in the Bush* can be accessed via the following link: https://www.alexismarieweaver.com/small-diffusion-works.html.

Both the personal listening space and the diffuse listening space are open to widespread abuse which determines spatial perception.

Therefore, perhaps it is more accurate to say that all performance spaces exist on a spectrum from the most NILS to the most ideal, dependent on a combination of environmental, personal and musical factors. The personal is entirely up to the listener, their level of engagement and ability to perceive the work. Similarly, the wealth of existing listening modes can be put into a series of spectra or grids which show both the listener's level of attention to a sound and "listening intention:" a subjective frame imposed by the listener, through which a sound object is perceived (Chion 1983). Michel Chion (1983) explains that Schaeffer's "reduced listening" is yet another kind of listening intention. Steven M Miller (2014) writes that, as humans, we switch between listening modes constantly. Chion's (1983) quadrant of listening modes are arranged as a "circuit" with each mode "involved simultaneously, interacting with each other." In the same way, listener immersion while in a NILS may fluctuate between almost total distraction from, and almost total focus on the musical work.

The following listening modes do not attempt to replace any of the wealth of existing listening modes outlined by other authors and practitioners, but to highlight environmental distraction as an important factor in the perception of a musical work.

4.1. Peripheral Listening

The composed work mingles with the ambient soundscape; the listener may be focused on another task or activity. Barry Truax (2001) describes "Background Listening" similarly: the sound is an expected element within the overall soundscape. Chion (1983) describes the "perceiving" listening mode as "the most elementary level…passive," whereas the "hearing" mode notices more important sounds, but does not necessarily infer anything from them. Peripheral Listening is more engaged than "perceiving" or "hearing;" examples might be an innocuous as radio or television in the background; music accompanying a religious ritual; or an ambient soundscape in a museum. Brian Eno's Ambient 1: Music for Airports (1978), described by Mark Richardson (2002) as "background enhancement," might fall into this category as an example of composing music for a non-ideal mental space – that is, a busy airport full of rushing and anxious people. It can be merely heard, it can be listened to, but adds something of value to the soundscape at all attention levels.

4.2. Central Listening

The listener intends for the composed work to have their undivided attention. Composers intending to eliminate the effects of the NILS attempt to draw the listener into this mode, falling in with most musical performances and works.

4.3. Dual Listening

The composed work fits into the real-world soundscape but exists as equal to the listener's attention, providing a framework for environmental sounds to fit in. Most likely a site-specific work, the composed work and environment are both the focus, working together to create the finished work. Sound Walks might fit into this category, often working with the visual environment to augment the listener's reality. Guitarist Sebastian Sequoiah-Grayson (2019) performed a series of live improvised guitar sets with ambient street noise broadcast into the performance venue; another example of how the ambient environment can contribute to, rather than distract from, the musical product.

These listening modes describe the level of engagement the listener has with the work, and the interaction of outside influences, or distractions. As discussed, small diffusion methods are likely to be used by listeners outside of a traditionally ideal listening space, causing elements of a NILS to impact upon the final listening mode of the listener. As such, these modes can be a useful framework for composers using small diffusion methods, allowing them to gauge the final listener's experience, and if necessary, adjust it. Depending on the specific technology used (which may be governed by the listener, depending on their agency), small diffusion can enhance the NILS for artistic purposes, or almost mask it completely. Therefore, the composer can use small diffusion to push listeners toward the listening mode they wish to engender.

5. 3 MINIATURES FOR HEADPHONES

My composition *3 miniatures for headphones* (2018)[1] aims to distract from distraction through its boisterous content and composing techniques. Like *Paranoia,* it therefore aims to incite a Central Listening experience. *3 miniatures* is a triptych composed for a specific NILS; in this instance, a concert space during interval. I knew the following of the space:

- It would be loud;
- It would be bright;
- Audience goers would be moving around the space, most likely conversing loudly.

3 miniatures were thus created to work within these parameters. Extremely soft dynamics and delicate textures were deemed inviable due to the danger of being inaudible. I decided to use attention-grabbing, or what I term "obnoxious," techniques such as hard left-right panning, strident frequencies and rapid switches between textural states. While I chose high-quality, close-backed headphones for this installation, the work was thus insured against the use of other non-noise cancelling headphones. A short, sharp structure was chosen to capture the audience's fleeting attention. Each miniature

[1] *3 miniatures for headphones* can be access via the following link: https://www.alexismarieweaver.com/small-diffusion-works.html.

is complementary, yet autonomous; the listener is free to listen to one or three before re-joining their companions.

The audience were instructed to use the provided headphones and press play on an old, early 2000s model CD player. The method of delivery and cheap burned CD together imparted a subtle crackling layer over the existing sound, adding a pleasant, unexpected depth to the texture which would not have happened if using a larger diffusion method.

6. CONCLUSION

It is crucial that modern composers of acousmatic music ask themselves, "where do I want my music to be listened to? By whom, and how?"

This paper does not strive to discount the vast opportunities for impact and immersion that large diffusion systems offer the listener. Nor does it strive to present definitive answers to the ongoing question of listener immersion and engagement across simple and complex diffusion systems. It does seek to open a robust discussion on the importance of the small diffusion space and the place they might hold in the future of acousmatic music.

Small diffusion and the NILS have been introduced as new terms to legitimise and encourage thought on the accessibility, portability and sustainability of new electroacoustic music. Composers may engage with one or both concepts, and still render their music more portable; considering the interaction of both can give the composer substantial control over the reception of their work outside of the traditional concert hall. The three introduced listening modes aim to assist the composer in choosing their desired interaction with (or mitigation of) what may be an unknown listening environment.

Moreover, these concepts encourage and challenge composers to create immersive music with what might be relatively simple or limited techniques. I have put forward two works as examples, with the hope that this will inspire more small diffusion works and a variety of interactions with the NILS. Composers may therefore harness the growing reach of playing devices to expand their listening community, and increase awareness and appreciation of the incredible genre of acousmatic music.

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